VNAV Lab 3: Modelling and Control of UAVs
Deliverable 2

Team Fullcans

A. Structure of quadrotors

We use a right-handed frame with x pointing right and y pointing up in the top-view exercise figure, and z pointing out of the page. This fixes all moment signs explicitly. The exercise figure and simulator figure use different rotor numbering; the matrices in this section follow the exercise figure, not the simulator figure. Let $d > 0$ be the centre-to-rotor distance and $a = d/\sqrt{2}$. The rotor positions in the exercise figure are

$$\rho_1 = (-a, a, 0)^T, \quad \rho_2 = (-a, -a, 0)^T, \quad \rho_3 = (a, -a, 0)^T, \quad \rho_4 = (a, a, 0)^T.$$

Define $u_i = \omega_i^2 \geq 0$, where ω_i is the speed magnitude in each propeller's configured lifting direction, $c_f > 0$ the lift coefficient, and $c_d > 0$ the reaction-torque coefficient. Then

$$f_i = c_f u_i, \qquad \rho_i \times (0, 0, f_i)^T = (\rho_{iy} f_i, -\rho_{ix} f_i, 0)^T.$$

Counter-clockwise rotor motion produces clockwise body reaction torque, so rotors 1 and 3 in (a) have negative yaw torque and rotors 2 and 4 have positive yaw torque. With $w = (f, M_x, M_y, M_z)^T$, we obtain

$$w = F_a u, \qquad F_a = \begin{bmatrix} c_f & c_f & c_f & c_f \\ ac_f & -ac_f & -ac_f & ac_f \\ ac_f & ac_f & -ac_f & -ac_f \\ -c_d & c_d & -c_d & c_d \end{bmatrix}. \tag{1}$$

The four scaled rows are mutually orthogonal and

$$\det F_a = 16a^2 c_f^3 c_d = 8d^2 c_f^3 c_d \neq 0, \qquad \operatorname{rank}(F_a) = 4.$$

Thus all four wrench components can be locally adjusted independently around a positive-speed hover, subject to actuator limits.

In configuration (b), swapping the propeller types and reversing the directions of rotors 3 and 4 preserves upward lift but reverses their yaw reaction torques. Therefore

$$F_b = \begin{bmatrix} c_f & c_f & c_f & c_f \\ ac_f & -ac_f & -ac_f & ac_f \\ ac_f & ac_f & -ac_f & -ac_f \\ -c_d & c_d & c_d & -c_d \end{bmatrix}. \tag{2}$$

Its fourth row is $-c_d/(ac_f)$ times its second row. The first three rows remain independent, hence

$$\boxed{\operatorname{rank}(F_b) = 3, \qquad M_z = -\frac{c_d}{ac_f} M_x.}$$

For example, $F_b(1, -1, 1, -1)^T = 0$. A pure yaw moment at zero roll moment is impossible. At an instant of level hover with zero angular velocity, keeping the position fixed requires zero roll acceleration and therefore $M_x = 0$, which also forces $M_z = 0$. A prescribed yaw acceleration at that instant cannot be realized. Consequently (b) cannot track an arbitrary independently prescribed output trajectory (x, y, z, Ψ); only trajectories compatible with the moment constraint may be feasible. Changing coordinate axes or wrench units cannot restore the missing rank.

B. Control with quadratic body drag

Let $R \in SO(3)$ rotate body vectors into the world frame, v be the world velocity, and $v^b = R^T v$. Write

$$D = \mathrm{diag}(0.1, 0.1, 0.2), \qquad h(v^b) = v^b \odot |v^b|, \qquad F_D^b = -Dh(v^b), \qquad F_D^w = RF_D^b.$$

Here $\odot$ means componentwise multiplication. This sign convention opposes motion in every axis and satisfies

$$(v^b)^T F_D^b = -0.1|v_x^b|^3 - 0.1|v_y^b|^3 - 0.2|v_z^b|^3 \le 0.$$

With the lab's upward z convention, the translational dynamics become

$$m\dot{v} = -mge_3 + fRe_3 + F_D^w, \qquad e_3 = (0,0,1)^T.$$

Define $e_x = x - x_d$, $e_v = v - v_d$ and command the rotor force vector

$$\boxed{A = -k_x e_x - k_v e_v + mge_3 + m\ddot{x}_d - \widehat{F}_D^w.} \tag{3}$$

Using measured velocity and attitude, compute $R^T v$, evaluate the quadratic drag law in the body frame, then rotate the result by R. Subtract this predicted world force when constructing A. For example, with $R = I$ and $v = (2,0,0)^T$, drag is $(-0.4, 0, 0)^T$ N, so the controller adds $+0.4$ N of forward force.

The compensated force changes both the desired thrust magnitude and attitude. For $\|A\| > 0$, choose

$$b_{3d} = \frac{A}{\|A\|}, \quad b_{1c} = (\cos\Psi_d, \sin\Psi_d, 0)^T, \quad b_{2d} = \frac{b_{3d} \times b_{1c}}{\|b_{3d} \times b_{1c}\|},$$

$$R_d = [\, b_{2d} \times b_{3d} \quad b_{2d} \quad b_{3d} \,], \qquad \boxed{f = A^T R e_3.}$$

With perfect attitude tracking, $fRe_3 = A$. If the drag estimate is exact, substitution into the dynamics gives

$$m\dot{e}_v = -k_x e_x - k_v e_v,$$

so the modelled disturbance is cancelled rather than left for feedback to counter through a persistent tracking error. With imperfect estimates, the residual disturbance is $F_D^w - \widehat{F}_D^w$.

The geometric attitude controller retains its form:

$$e_R = \tfrac{1}{2}(R_d^T R - R^T R_d)^\vee, \quad e_\Omega = \Omega - R^T R_d \Omega_d,$$

$$M = -k_R e_R - k_\Omega e_\Omega + \Omega \times J\Omega - J(\widehat{\Omega} R^T R_d \Omega_d - R^T R_d \dot{\Omega}_d).$$

If the full feedforward terms are used, Ω_d and $\dot{\Omega}_d$ must come from the compensated R_d. The drag model supplied in the exercise is a force at the centre of mass; it does not specify an extra body moment.

In the paper's downward z convention, instead use $A_p = -k_x e_x - k_v e_v - mge_3 + m\ddot{x}_d - \widehat{F}_D^w$, $b_{3d} = -A_p/\|A_p\|$ and $f = -A_p^T R e_3$. The implementation provides optional drag compensation, disabled by default because the hypothetical exercise drag need not match the TESSE plant.

References

1. MIT VNAV, *Lab 3 Exercises (2023)*, `https://vnav.mit.edu/labs_2023/lab3/exercises.html`.

2. T. Lee, M. Leok, and N. H. McClamroch, *Geometric Tracking Control of a Quadrotor UAV on SE(3)*, CDC 2010, pp. 5420–5425, `https://mleok.science/pdf/LeLeMc2010_quadrotor.pdf`.